Relevant Papers on Edelstein Effect for Rashba Fermions

1. **Edelstein Effect in Isotropic and Anisotropic Rashba Models** (Most Relevant - Very Recent)

- **arXiv ID:** 2503.20712
- **Authors:** Irene Gaiardoni, Mattia Trama, Alfonso Maiellaro, Claudio Guarcello, Francesco Romeo, Roberta Citro
- **Date:** 2025-03-26
- **URL:** https://arxiv.org/abs/2503.20712
- **PDF:** https://arxiv.org/pdf/2503.20712v1
- **Summary:** This paper investigates spin-to-charge conversion via the Edelstein effect in a 2D Rashba electron gas using the semiclassical Boltzmann approach. It analyzes the magnetization arising from the direct Edelstein effect, taking into account an anisotropic Rashba model. The study explicitly examines how the effect depends on the effective parameters including spin-orbit coupling strength, Fermi velocity, and chirality. The work provides analytical expressions for magnetization magnitude and direction under different electric field configurations, making it ideal for building a computational model with explicit graphics showing parameter dependencies.

2. **Spin and Orbital Edelstein Effect in Spin-Orbit Coupled Noncentrosymmetric Superconductor** (Very Recent)

- **arXiv ID:** 2408.08151
- **Authors:** Satoshi Ando, Yukio Tanaka, Mario Cuoco, Luca Chirolli, Maria Teresa Mercaldo
- **Date:** 2024-08-15
- **URL:** https://arxiv.org/abs/2408.08151
- **PDF:** https://arxiv.org/pdf/2408.08151v2
- **Summary:** This recent work focuses on understanding the relation between spin and orbital properties of Edelstein effects in conventional spin-singlet noncentrosymmetric superconductors. The paper derives explicit formulas for current-induced magnetization in

systems with Rashba-type spin-orbit coupling. It includes detailed analysis of how magnetization direction and magnitude vary with applied electric field direction and magnitude, and discusses the role of Fermi surface topology and spin-orbit coupling strength. The paper contains numerical results and graphics showing parameter dependencies that can be directly adapted for Rashba fermion models at the Gamma point.

3. **Spin and Orbital Edelstein Effect in a Bilayer System with Rashba Interaction** (Recent)

- **arXiv ID:** 2307.02872
- **Authors:** Sergio Leiva M., Jürgen Henk, Ingrid Mertig, Annika Johansson
- **Date:** 2023-07-06
- **URL:** https://arxiv.org/abs/2307.02872
- **PDF:** https://arxiv.org/pdf/2307.02872v2
- **Summary:** This paper presents a comprehensive study of the spin Edelstein effect and the recently predicted orbital Edelstein effect in bilayer systems with Rashba spin-orbit coupling. The authors develop a theoretical framework to calculate current-induced spin and orbital magnetization, providing explicit expressions for magnetization as a function of electric field strength and direction. The work includes detailed discussions on how the effect depends on spin-orbit coupling strength, Fermi energy, and chirality of the bands. The paper contains multiple graphics showing the angular dependence of magnetization and parameter sweeps that are directly applicable to building a model for Rashba fermions at the Gamma point of the Brillouin zone.

Key Features Across All Papers for Model Building:

Feature	Paper 1 (2503.20712)	Paper 2 (2408.08151)	Paper 3 (2307.02872)
Rashba Model at Γ-point	✓ Explicit	✓ Implicit	✓ Explicit
E-field Direction Dependence	✓	✓	✓
E-field Magnitude Dependence	✓	✓	✓

Feature	Paper 1 (2503.20712)	Paper 2 (2408.08151)	Paper 3 (2307.02872)
SOC Strength Parameter	✓	✓	✓
Fermi Velocity Dependence	✓	✓	✓
Chirality Effects	✓	✓	✓
Explicit Graphics	✓	✓	✓

These three papers provide comprehensive theoretical frameworks, analytical expressions, and numerical results that will enable the construction of a complete model for calculating the Edelstein effect in Rashba fermion systems with explicit visualization of parameter dependencies.